Formula sheet

The equations an examiner might ask you to write, each with a plain-English reading.

What	Formula	Read it as
Bloch wave	$H(\mathbf{r}) = e^{i\,\mathbf{k}\cdot\mathbf{r}}\,u(\mathbf{r})$, u repeats in every cell	A wave in a crystal is a plane wave times a pattern that repeats every cell; k sets the phase step from cell to cell.
Master equation (2D, TE)	$-\cdot((1/\varepsilon(\mathbf{r})) H) = (\omega/c)^2 H$	How the field curves, weighted by 1/ε, equals frequency squared times the field: an eigenvalue problem.
Plane-wave expansion	$\Sigma_{G'} (\mathbf{k}+\mathbf{G})\cdot(\mathbf{k}+\mathbf{G'})\, \eta_{G,G'}\, h_{G'} = (\omega/c)^2\, h_G$	Writing H and ε as sums of plane waves turns the equation into a matrix problem M·h = (ω/c)² h; η is the inverted matrix of ε's Fourier numbers (Ho's rule). Eigenvalues give the band frequencies.
Frequency from eigenvalue	$f = c \cdot \sqrt{\lambda} / (2\pi)$	λ = (ω/c)² is the eigenvalue; c = 299,792 km/s.
Band edges	f_{lower} = max(band 1 at M, K), f_{upper} = min(band 2 at M, K), gap = $f_{upper} - f_{lower}$	Verified: the extremes always sit at M or K for these crystals (within 0.003 GHz).
Hole shape	$r(\theta) = r_0 + c_3 \cos 3\theta + s_3 \sin 3\theta + c_6 \cos 6\theta + s_6 \sin 6\theta$	5 numbers per hole, 10 per cell; only multiples of 3 keep the 120° symmetry.
Neuron	output = SiLU(w·x + b), $\mathrm{SiLU}(z) = z / (1 + e^{-z})$	Each layer: weighted sum plus bias, then a smooth bend.
Tandem loss	$L = \Sigma_{edges} ((f_{pred} - f_{target})/\sigma)^2 + \lambda\cdot\mathrm{penalty}(p)$, λ = 0.001	Match the target through the frozen forward network; the penalty keeps walls and radii fabricable.
CVAE loss	$L = \|\|u - \hat{u}\|\|^2/0.01 + \beta\cdot KL(N(\mu, \sigma^2) \,\|\|\, N(0, 1))$, β = 0.5	Rebuild the design well, while keeping the codes z close to a standard normal so random z give valid designs.
Multi-fidelity correction	$f_{COMSOL}(x) \approx f_{PWE}(x) + \Delta(x)$, Δ ~ Gaussian process	The GP learns the small, smooth difference between the two solvers from COMSOL cells.
Accuracy	MAE = mean $\|f_{pred} - f_{true}\|$; design error = max(|Δf_lower|, |Δf_upper|) < 1 GHz	Typical error per edge in GHz; a design counts only if its worse edge is within 1 GHz.
Scaling	f 1/a	Scaling the whole crystal by s divides every frequency by s (2D, non-dispersive silicon).